```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
```

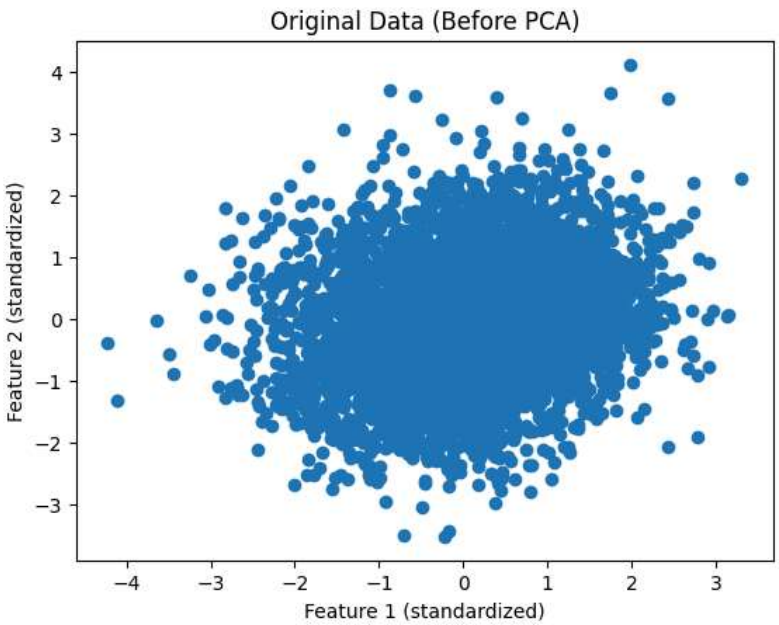
```
df = pd.read_csv("PCA_5D_dataset.csv")
```

```
X = df.select_dtypes(include=["number"])
```

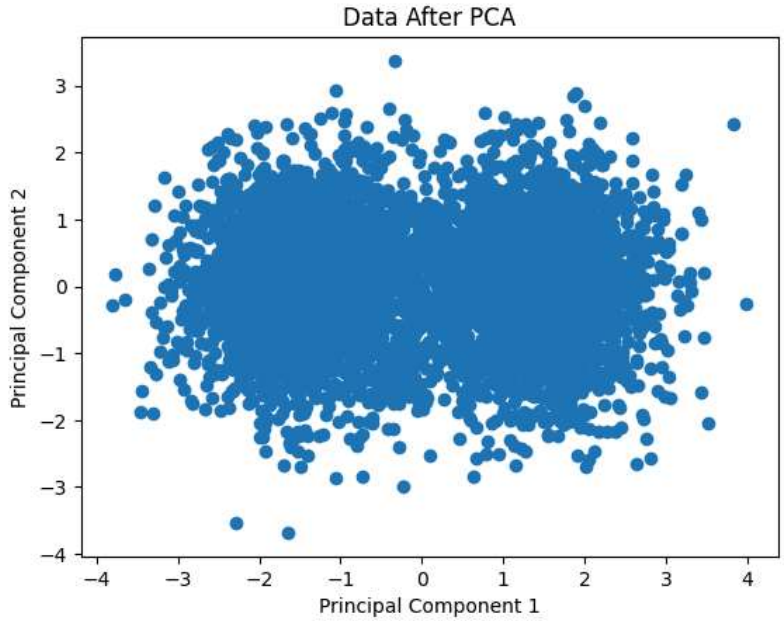
```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
```

```
plt.figure()
plt.scatter(X_scaled[:, 0], X_scaled[:, 1])
plt.xlabel("Feature 1 (standardized)")
plt.ylabel("Feature 2 (standardized)")
plt.title("Original Data (Before PCA)")
plt.show()
```




```
plt.figure()
plt.scatter(X_pca[:, 0], X_pca[:, 1])
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("Data After PCA")
plt.show()
```




```
print("Explained Variance Ratio:", pca.explained_variance_ratio_)
```

```
Explained Variance Ratio: [0.4136642  0.13733514]
```